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Studierichting: Informatica
Oefeningenreeks 2B nr. 5.3

$$\text{Bgtan}(x) + \text{Bgtan}\left(\frac{1}{x}\right)$$

$$\text{Stel } x = \tan(\alpha) \text{ met } \alpha \in \left]0, \frac{\pi}{2}\right[$$

$$\Rightarrow \frac{1}{x} = \cot(\alpha) = \tan\left(\frac{\pi}{2} - \alpha\right)$$

$$\text{Dus } \text{Bgtan}\left(\frac{1}{x}\right) = \frac{\pi}{2} - \alpha$$

$$\Rightarrow \text{Bgtan}(x) + \text{Bgtan}\left(\frac{1}{x}\right) = \alpha + \left(\frac{\pi}{2} - \alpha\right) = \frac{\pi}{2}$$

Conclusie:

$$\forall x > 0 : \text{Bgtan}(x) + \text{Bgtan}\left(\frac{1}{x}\right) = \frac{\pi}{2}$$

References